

Reflection Capture for Graph CCR Algebras

Run fa_banach_001, lane 19

June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the von Neumann algebra classification, it asks for a noncommutative C^* -algebra version of the commutative $C(K)$ BCP/UBCP criterion:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

Theorem 1 (reflection-capture criterion). *Let A be a C^* -algebra and let $(u_v)_{v \in V} \subset S_A$, where V is uncountable. Suppose:*

1. *for every countable $C \subset A$, there is a countable $S \subset V$ such that $C \subset C^*(u_s : s \in S)$;*
2. *for every $v \in V$, there is an isometric automorphism $\alpha_v \in \text{Aut}(A)$ satisfying*

$$\alpha_v(u_v) = -u_v, \quad \alpha_v(u_w) = u_w \quad (w \neq v).$$

Then A has countable capture. Consequently A fails BCP.

Here countable capture means that for every countable $C \subset A$ there is $u \in S_A$ such that

$$\text{dist}(c, \mathbb{C}u) = \|c\| \quad (c \in C).$$

The packet

2311.14257_countable_capture_characterization

records that countable capture implies failure of BCP.

Corollary 1. *Let $G = (V, E)$ be an uncountable graph and let $B(G)$ be Farah's graph CCR algebra generated by self-adjoint unitaries $(u_v)_{v \in V}$ with*

$$u_v u_w = u_w u_v \quad \text{if } \{v, w\} \notin E, \quad u_v u_w = -u_w u_v \quad \text{if } \{v, w\} \in E.$$

Then $B(G)$ has countable capture and fails BCP.

Corollary 2. *The simple nuclear AM algebra of density continuum constructed by Farah in [2] has countable capture and fails BCP.*

Idea of the Proof

The missing mechanism in the arbitrary simple case is an exact way to add one new direction over a countable old subalgebra. For graph CCR algebras the new direction does not have to commute tensorially with the old algebra. It is enough that an automorphism fixes the old algebra and flips the new generator.

Indeed, if $\alpha(c) = c$ and $\alpha(u) = -u$, then

$$\|c - \lambda u\| = \|\alpha(c - \lambda u)\| = \|c + \lambda u\|.$$

Since c is the average of $c - \lambda u$ and $c + \lambda u$, convexity forces

$$\|c\| \leq \|c - \lambda u\|$$

for every scalar λ . This is exactly the countable-capture distance inequality.

Proof of the Criterion

Proof of Theorem 1. Let $C \subset A$ be countable. By hypothesis, choose a countable $S \subset V$ such that $C \subset C^*(u_s : s \in S)$. Since V is uncountable, choose $v_0 \in V \setminus S$, and set $u = u_{v_0}$.

The automorphism $\alpha = \alpha_{v_0}$ fixes every generator u_s with $s \in S$. Therefore it fixes every element of $C^*(u_s : s \in S)$, and in particular

$$\alpha(c) = c \quad (c \in C).$$

It also satisfies $\alpha(u) = -u$.

Fix $c \in C$ and $\lambda \in \mathbb{C}$. Since α is isometric,

$$\|c - \lambda u\| = \|\alpha(c - \lambda u)\| = \|c + \lambda u\|.$$

But

$$c = \frac{1}{2}(c - \lambda u) + \frac{1}{2}(c + \lambda u),$$

and hence

$$\|c\| \leq \frac{1}{2}\|c - \lambda u\| + \frac{1}{2}\|c + \lambda u\| = \|c - \lambda u\|.$$

Taking the infimum over $\lambda \in \mathbb{C}$ gives

$$\text{dist}(c, \mathbb{C}u) \geq \|c\|.$$

The reverse inequality is obtained by taking $\lambda = 0$. Thus $\text{dist}(c, \mathbb{C}u) = \|c\|$ for every $c \in C$. This proves countable capture. Countable capture implies failure of BCP by the previously packaged countable-capture obstruction. $\square$

Graph CCR Algebras

We recall only the features of Farah's construction needed here. For a graph $G = (V, E)$, Farah defines $B(G)$ as the universal C^* -algebra generated by self-adjoint unitaries $(u_v)_{v \in V}$ satisfying the commutation/anticommutation relations displayed in Corollary 1. Farah proves that $B(G)$ is well-defined for every graph G , and that for an infinite graph its character density is $|V|$ [2, Lemma 1.1].

Lemma 1. *If $G = (V, E)$ is a graph and $C \subset B(G)$ is countable, then there is a countable $S \subset V$ such that*

$$C \subset C^*(u_s : s \in S).$$

Proof. Farah's construction realizes $B(G)$ as the direct limit of the algebras $B(G_0)$, where G_0 ranges over finite induced subgraphs of G . Thus the union of the finite-vertex subalgebras

$$C^*(u_s : s \in F), \quad F \subset V \text{ finite},$$

is dense in $B(G)$.

For each $c \in C$, choose a sequence of finite-vertex algebra elements converging to c . Let S_c be the union of all finite vertex sets used in this sequence. Then S_c is countable and

$$c \in C^*(u_s : s \in S_c).$$

Putting $S = \bigcup_{c \in C} S_c$, we obtain a countable set of vertices with $C \subset C^*(u_s : s \in S)$. □

Lemma 2. *For each $v \in V$, the assignment*

$$u_v \mapsto -u_v, \quad u_w \mapsto u_w \quad (w \neq v)$$

extends to an automorphism of $B(G)$.

Proof. The modified family of self-adjoint unitaries satisfies exactly the same relations as the original family: multiplying one generator by -1 does not change whether it commutes or anticommutes with any other generator. By the universal property of $B(G)$, the assignment therefore induces a unital $*$ -homomorphism $B(G) \rightarrow B(G)$. Applying the same assignment twice gives the identity on generators, hence the homomorphism is an automorphism and is isometric. □

Proof of Corollary 1. For an uncountable graph $G = (V, E)$, Lemma 1 verifies the countable-support hypothesis in Theorem 1, and Lemma 2 verifies the reflection hypothesis. The theorem gives countable capture, hence failure of BCP. □

Proof of Corollary 2. Farah constructs an independent dense family $\mathcal{A} \subseteq 2^{\mathbb{N}}$ of cardinality $2^{\aleph_0}$ and studies the bipartite graph

$$G(\mathbb{N}, \mathcal{A}).$$

He proves that $B(G(\mathbb{N}, \mathcal{A}))$ is simple, nuclear, unital, AM, has unique trace, has density continuum, and has irreducible representations on both a separable and a nonseparable Hilbert space [2, Theorem 1 and Lemmas 1.4, 1.6, 1.8]. Since the vertex set $\mathbb{N} \cup \mathcal{A}$ is uncountable, Corollary 1 applies and proves that this simple nuclear density-continuum algebra has countable capture and fails BCP. □

Verification Notes and Limitations

The proof is exact: no approximate centrality, saturation, or limiting compactness is used after choosing the unused generator. The only structural ingredients are countable support and a sign-flip automorphism.

This substantially narrows the live simple density-continuum wall because it settles a canonical nonseparable simple nuclear example with a faithful separable representation. It does not settle arbitrary simple nonseparable C^* -algebras. A remaining example would have to lack this reflection symmetry, as well as the previously packaged tensor, product, group, von Neumann, primitive-topological, corona, and saturation mechanisms.

Novelty and Search Bounds

Local indexes were searched for arXiv:2311.14257 together with graph CCR, Farah, self-adjoint unitaries, relative commutants, simple nonseparable C^* -algebras, BCP, and countable capture. Web searches on June 17, 2026 for graph CCR algebras plus ball-covering and capture terms did not find a prior BCP statement of this reflection mechanism. The Farah paper is used for the construction and structural properties of the application; the countable-capture proof above is self-contained once the graph CCR universal presentation is available.

Human Review Focus

Reviewers should check:

1. the equality $\|c - \lambda u\| = \|c + \lambda u\|$ from the sign-flip automorphism;
2. the convexity step proving $\|c\| \leq \|c - \lambda u\|$;
3. the countable-support lemma for direct limits over finite subgraphs;
4. that sign flips preserve the graph CCR relations;
5. the cited Farah hypotheses for the simple nuclear AM density continuum application.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Ilijas Farah, *Graphs and CCR algebras*, arXiv:0908.3927, 2009.